Frame-Relative Counterfactual Necessity for Explaining Synthesized Temporal Behavior

Abstract

LTL synthesis is many-to-one: a controller alone fixes neither a unique specification nor a unique temporal explanation of its behavior. We study the reverse question under a declared *causal frame*: given background, intervention model, candidate vocabulary, similarity order, and preference order, when is a candidate C a *frame-relative counterfactual necessary condition* for effect E?

We separate two notions throughout. **Trace-existential necessity (N1)** applies to a Mealy machine M, trace τ, and frame: deciding validity for a fixed C is a one-player shortest-edit search plus non-emptiness check (the frozen machine has no free move). N1 is confined to co-safety / finite-change causes; ω-liveness commitments require the strategy level. **Strategy-guarantee necessity (N2)** additionally requires a synthesis artifact $(G_\varphi, \sigma, V_{\mathfrak{F}})$ (not determined by M alone): a constrained *reverse parity game* decides whether the controller can drop C and still guarantee E against admissible environments (parity winning region, then a one-player check that re-imposes the objective).

For a *fixed* candidate C and finite candidate space $\mathcal{K}_{\mathfrak{F}}$, validity and $\lhd$-minimality are decidable under stated bounds; we do *not* present a complete solver or a tight complexity theorem. Canonical default frames fix every component except the declared vocabulary, grammar, and tie-breaks; the answer is then a function of (M, τ, E) *given that declaration.* A finite-trace specialization aligns with TempoBench-style TCE *conditionally* on machine-checkable assumptions: a control-validated audit confirms them on the publicly available TempoBench sample artifact, and a reference implementation of the finite construction reproduces five worked instances from HOA encodings and matches the one available TempoBench answer key per cell; the grader is profiled (sub-second full-key computation through small windows, with the actual-causation mode the measured exponential), and on a 107-instance random-transducer corpus the per-cell actual-cause and determining-set unions coincided throughout. Corpus-scale evaluation on real synthesized artifacts awaits the dataset's re-release.

Contents

1 Introduction

The standard synthesis pipeline is

$$\varphi_{\mathrm{LTL}} \longrightarrow A_\varphi \longrightarrow G_\varphi \longrightarrow \sigma \longrightarrow M,$$

where φ_{LTL} is an LTL specification, A_φ an automaton for its models, G_φ a parity game, σ a winning strategy, and $M = (S, s_0, I, O, \delta, \lambda)$ a Mealy implementation ($\delta : S \times 2^I \to S$, $\lambda : S \times 2^I \to 2^O$). The pipeline loses information: many specifications yield the same machine and many strategies are behaviorally equivalent, so the reconstructive reverse map $M \to \varphi$ is ill-posed.

We pose instead an *explanatory* reverse question. The inputs split by regime. **Trace-level analysis** takes $(M, \mathfrak{F}, E, \tau)$: a Mealy machine $M = (S, s_0, I, O, \delta, \lambda)$ fixing the I/O partition ($I \cap O = \emptyset$), a causal frame $\mathfrak{F}$, a temporal effect E (a single, possibly conjunctive, ω-regular trace property), and observed trace $\tau \in \mathcal{L}(M)$. **Strategy-level analysis** additionally requires a synthesis/game artifact $(G_\varphi, \sigma, V_{\mathfrak{F}})$ — parity game $G_\varphi = (V, V_0, V_1, T, \Omega)$, frozen winning strategy σ implemented by M, and editable vertex set $V_{\mathfrak{F}} \subseteq V_0$ — which M alone does not determine. Optional specification φ may justify G_φ and $\mathfrak{F} = \mathrm{Frame}(\varphi; \eta)$, but the strategy query is not a function of (M, τ, E) alone. Throughout we use "cause" for a *frame-relative necessary condition*; this is counterfactual necessity, not Halpern–Pearl actual causation (Section 7).

The task is to *define and check* candidates and, when $\mathcal{K}_{\mathfrak{F}}$ is finite or bounded, to *search* for a $\lhd$-minimal valid one $C^\star$; we do not claim a complete solver for open LTL candidate spaces or minimal dominion extraction. A trace is the infinite word $\tau = a_0 a_1 \cdots$ with $a_t = i_t \cup o_t \in 2^{I \cup O}$, $o_t = \lambda(s_t, i_t)$, $s_{t+1} = \delta(s_t, i_t)$; in implementations τ is a lasso uv^ω.

The intended output is not "the controller wins," but a temporal explanation of what had to remain true for the chosen effect to occur, *relative to a declared frame*. We commit to a *counterfactual-necessity* reading: C holds on τ, removing C is realizable under the frame, and on the closest admissible executions that remove C the effect fails. This is not sufficiency and not Halpern–Pearl actual causation (which adds a minimal-sufficient-set requirement); Section 7 discusses what the necessity reading excludes. We avoid the unqualified phrase "weakest necessary condition": in standard logic the *strongest* necessary condition for E (equivalently the weakest *sufficient* condition) is E itself, while the weakest necessary condition is true, so the phrase is doubly misleading for what we do. Instead, "more permissive" means larger language $\mathcal{L}(C)$, "insufficient" means "does not by itself force the effect," and these are kept distinct.

2 Related Work and Contributions

This work sits at the intersection of three lines of research; our claims are deltas relative to them.

Counterfactual and actual causation. The necessity core is Lewis–Stalnaker: a cause is what, on the closest worlds where it is absent, removes the effect, with centering and a limit assumption on the similarity order. Halpern–Pearl actual causation adds a minimal-sufficient-set requirement and a contingency quantifier; we adopt only the necessity half (Section 7) and are explicit about what this excludes (preemption discrimination, sufficiency).

Temporal/ω-regular causality. The closest technical relatives are automata-theoretic accounts of causes for ω-regular effects in reactive systems (downward-closed ω-regular causes via game/automaton constructions), which also underlie TempoBench ground truth. Our delta is not "another parity solve" but *what the game is asked*:

- **Centering:** N1 is trace-existential (closest admissible counterfactual trace); N2 is strategy-centered (can σ drop a commitment while still guaranteeing E?), without Lewis–Stalnaker trace centering.
- **Objective:** $B_{\mathfrak{F}} \to E$ with non-commitment witnessed on $B_{\mathfrak{F}} \wedge \neg C$ plays, not a single global ω-regular cause language.
- **Intervention:** re-opening a declared vertex subset $V_{\mathfrak{F}}$ of a *frozen* σ, not synthesizing an unconstrained alternative controller.
- **Framing:** background, admissible counterfactuals, similarity, candidate vocabulary, and anti-circularity are explicit parameters; prior constructions fix many of these implicitly.

We do not claim a new causality-synthesis algorithm; the contribution is this parameterization plus sound decision procedures for fixed C (and bounded search when $|\mathcal{K}_{\mathfrak{F}}|$ is finite).

Reactive synthesis and parity games. We use only standard facts: LTL→deterministic parity automaton translation; parity games are determined, solvable in quasi-polynomial time, and lie in UP $\cap$ co-UP; HOA is the interchange format for ω-automata. These are background.

Contributions. (1) An explicit *causal frame* $\mathfrak{F}$ factoring a temporal causal query into background, effect scope, intervention model, candidate vocabulary, similarity order, and preference order, with canonical default frames (Section 4.8) that pin all components but the declared vocabulary, grammar, and tie-breaks. (2) A clean separation of trace-existential necessity (N1: one-player shortest-edit + non-emptiness) from strategy-guarantee necessity (N2: reverse parity game with sound-and-complete decision for fixed C, Section 6). (3) A decidable anti-circularity condition (Assumption 2) — semantic position-masking for point/bounded effects, pure-logic non-entailment for all effects, and a mandatory effect cone for general ω-regular effects — that bars logical entailment and point-position restatement while admitting input/invariant causes. (4) A finite-trace specialization aligning with TempoBench-style TCE *conditionally* on (F1)–(F2), which we have not verified (Section 9). Core deliverables are the definitions and Propositions 3–4; a tight complexity theorem, dominion minimization, and empirical evaluation are left open.

Scope (read once). Established: valid-cause clauses; N1/N2 distinction; Proposition 3 (N2 decision for fixed $V_{\mathfrak{F}}, C$); certificate characterization (Proposition 2); finite specialization (Proposition 4). **Scoped:** candidate checking and search over finite/bounded $\mathcal{K}_{\mathfrak{F}}$; conditional complexity (Section 8); TempoBench alignment on (F1)–(F2). **Not claimed:** full solver, HP-style actual causation, verified benchmark artifacts, or tight minimal-dominion completeness.

3 Background: Parity Games

A parity game is $G = (V, V_0, V_1, T, \Omega)$ with vertices $V = V_0 \sqcup V_1$ (system V_0, environment V_1), a total edge relation $T \subseteq V \times V$, and a priority map $\Omega : V \to \mathbb{N}$. A play is an infinite path; under the *min-even* convention used throughout, the system wins iff the least priority seen infinitely often is even. LTL formulas translate to deterministic parity automata, so synthesis reduces to deciding $\exists\sigma\,\forall\rho.\ \mathrm{Play}(\sigma, \rho) \models \varphi$; a winning σ is implementable as a Mealy machine. We reuse this machinery in reverse: we freeze M as the strategy and ask which temporal conditions were necessary for a chosen effect.

4 The Formal Framework

4.1 The causal frame and admissible counterfactuals

The reverse-causal object is the frame

$$\mathfrak{F} = (B_{\mathfrak{F}}, E_{\mathfrak{F}}, S_{\mathfrak{F}}, O_{\mathfrak{F}}, \mathcal{K}_{\mathfrak{F}}, \mathcal{I}_{\mathfrak{F}}, \preceq, \lhd),$$

where $B_{\mathfrak{F}}$ is a background constraint (environment assumptions and system guarantees, when present, are folded in; $B_{\mathfrak{F}} = \mathsf{true}$ if none), $E_{\mathfrak{F}} = \bigwedge_{r \in S_{\mathfrak{F}}} E_r$ is the effect selected from a decomposition $E = E_1 \wedge \cdots \wedge E_m$ by indices $S_{\mathfrak{F}}$ (if no scoping, $E_{\mathfrak{F}} = E$), $O_{\mathfrak{F}} \subseteq O$ the relevant outputs, $\mathcal{K}_{\mathfrak{F}}$ the candidate-cause space, $\mathcal{I}_{\mathfrak{F}}$ the intervention model, $\preceq\ = \{\preceq_\tau\}_\tau$ the similarity orders, and $\lhd$ the candidate-preference order ($C' \lhd C$: C' is preferred). When derived from a specification we write $\mathfrak{F} = \mathrm{Frame}(\varphi; \eta)$; $\mathfrak{F}$ is *not* φ and is unrelated to a Kripke frame. A useful slogan: *M gives behavior; $\mathfrak{F}$ gives explanatory scope.*

The intervention model fixes the admissible counterfactual set $\mathrm{Adm}_M^{\mathfrak{F}}(\tau)$ in one of three regimes:

Input-only

counterfactuals stay genuine machine behaviors, $\mathrm{Adm}_M^{\mathfrak{F}}(\tau) = \mathcal{L}(M) \cap \mathcal{L}(B_{\mathfrak{F}})$. (Default; asks which environment behavior caused the effect.)

Output-intervention

outputs may be edited away from λ: $\mathrm{Adm}_M^{\mathfrak{F}}(\tau) = \mathrm{Edit}_O^{\mathfrak{F}}(\mathcal{L}(M)) \cap \mathcal{L}(B_{\mathfrak{F}})$, where $\mathrm{Edit}_O^{\mathfrak{F}}$ toggles outputs at a frame-permitted position set; a separate output/event-causality regime.

Strategy-structure

the system's commitments are re-opened: $\mathrm{Adm}_M^{\mathfrak{F}}(\tau) = \{\mathrm{Play}(\sigma', \rho) \mid \sigma' \in \Sigma_{\mathfrak{F}},\ \rho\} \cap \mathcal{L}(B_{\mathfrak{F}})$, with

$$\Sigma_{\mathfrak{F}} = \{\sigma' \mid \sigma' \text{ agrees with } \sigma \text{ on } V_0 \setminus V_{\mathfrak{F}}\}$$

for a declared editable set $V_{\mathfrak{F}} \subseteq V_0$ of the synthesis game $G_\varphi = (V, V_0, V_1, T, \Omega)$ and frozen strategy σ (we require $\sigma \in \Sigma_{\mathfrak{F}}$). By default we take σ' to range over *finite-memory* strategies on G_φ (sufficient for ω-regular objectives on finite arenas); the witness search returns an accepting lasso π on a finite sub-arena, from which such a σ' exists by standard synthesis (Section 6). This regime requires inputs $(G_\varphi, \sigma, V_{\mathfrak{F}})$ in addition to M.

The τ-argument is kept uniformly even though the first two regimes do not depend on τ, because frame-relative interventions (the finite-horizon TempoBench specialization of Section 9) perturb τ itself.

4.2 Similarity to the actual trace

Counterfactual causality needs a notion of closeness, a preorder $\preceq_\tau$ where $\tau' \preceq_\tau \tau''$ reads "τ' is at least as close to τ as τ''." Our canonical order is *count-first*: let

$$\mathrm{Diff}_\tau(\tau') = \{t \mid \tau'(t) \neq \tau(t)\}, \quad d(\tau, \tau') = |\mathrm{Diff}_\tau(\tau')| \in \mathbb{N} \cup \{\infty\},$$

and define the primary key by d alone (fewer changed positions is closer; ∞ on top). Traces with an *infinite* change set form a single $\preceq_\tau$-maximal class we do not order internally; Close (below) is taken over *finite-change* removals whenever any exist. Under this *count-only* order, $\mathrm{Close}^{\mathfrak{F}}_{\tau,M}(\neg C)$ is the full set of minimum-count admissible $\neg C$-removals, and validity quantifies over *all* of them (effect must fail on each). This matches minimal-intervention practice (Hamming distance on traces) and avoids an arbitrary tie-break changing validity when two minimum-count removals differ only in *which* positions changed.

Definition 1 (Optional profile refinement). *A frame may declare a total order $\lhd_\Sigma$ on $2^{I\cup O}$ and refine $\preceq_\tau$ lexicographically by difference profile* prof_τ *(changed positions in order, each tagged with its new value). On finite-change traces this yields a* total *order with unique minimum τ (centering) and unique* Close *(a single witness). Validity then depends on that tie-break; reports must state $\lhd_\Sigma$ or use count-only (default in $\mathfrak{F}_{\mathrm{trace}}$ and $\mathfrak{F}_{\mathrm{TB}}$).*

On finite-change traces the count-only order is antisymmetric among distinct traces at fixed d, well-founded (the count key is well-ordered), and has unique minimum τ among admissible traces. Lexicographic comparison of *infinite*-change profiles is *not* well-founded, which is why Close is restricted to finite-change removals; if the only removals require infinitely many edits, Close is empty and the candidate fails non-vacuity. This is a **scope boundary**, stated upfront: trace-level necessity (N1) is confined to *co-safety / finite-change* causes. An ω-liveness commitment whose removal requires infinitely many trace edits (e.g. removing $GF\,req$ from a lasso forces $FG\,\neg req$) has Close $= \emptyset$ at the trace level and belongs to N2, where "dropping a commitment" is one strategic deviation (Section 6). Neither N1 nor N2 alone is a complete account of temporal explanation; the split reflects an inherent limit of trace-level Lewis–Stalnaker necessity for liveness.

We reject two common alternatives on principled grounds: the first-difference (Baire) order ties traces that differ only in *when* the same minimal edit occurs, and pure change-set *inclusion* makes incomparable edits at the same count — both blur "minimal intervention" when several same-count removals exist.

Remark 1 (Closest removals for lasso counterfactuals). When $\tau = uv^\omega$, M is finite-state, and the regime is input-only or output-intervention, every admissible $\neg C$-trace is a bounded-period lasso (a path in the finite product $M \times A_{\neg C} \times A_E$), and d is finite exactly on perturbations agreeing with τ on its tail. Under count-only $\preceq_\tau$, $\mathrm{Close}^{\mathfrak{F}}_{\tau,M}(\neg C)$ is the set of minimum-count removals, found by BFS over that product; under the optional profile refinement (Definition 1) it is a unique shortest-edit witness.

Remark 2 (Centering and the strategy-structure regime). Strategy-guarantee necessity (N2, Section 4.6) is a *type-level* property of the controller σ — "can σ drop a commitment and still guarantee E?" — and is centered on the actual *strategy* σ, not on the actual trace τ. It therefore does *not* require trace-centering (Assumption 1(ii)), and Proposition 3 does not use it; the trace-level Lewis–Stalnaker axiom is simply not the right notion at the strategy level. Trace-centering matters only for the *token-level* certificate $C^\flat$ (Proposition 2): if one wishes to run that construction *within* a strategy-structure frame (a separate, optional use), the edited-vertex subset order ties all plays of σ with τ, so one must refine it lexicographically (edited-vertex subset, then the trace-level count-first order) to restore τ as unique minimum. Absent that refinement, Proposition 2 is read only for the input-only/output regimes, while the strategy-structure regime carries the validity content of Proposition 3 (which needs no centering).

Assumption 1 (Well-formedness). *Fix $M, \mathfrak{F}, \tau$. (i)* ***Actual-world admissibility:*** $\tau \in \mathrm{Adm}^{\mathfrak{F}}_M(\tau)$. *(ii)* ***Centering:*** *$\preceq_\tau$ has τ as unique minimum ($\tau \prec_\tau \tau'$ for admissible $\tau' \neq \tau$). (iii)* ***Well-foundedness (Limit Assumption):*** *$\preceq_\tau$ has no infinite strictly descending chain on $\mathrm{Adm}^{\mathfrak{F}}_M(\tau)$, so every non-empty admissible set has minimal elements and every element dominates a minimal one. Parts (i)–(ii) are Lewis–Stalnaker centering; (iii) holds for the canonical order when* Close *is restricted to finite-change removals (Section 4.2), and is otherwise an explicit hypothesis on $\preceq_\tau$.*

4.3 Valid causes

The closest admissible counterfactuals removing C are

$$\mathrm{Close}^{\mathfrak{F}}_{\tau,M}(\neg C) = \min_{\preceq_\tau}\{\tau' \in \mathrm{Adm}^{\mathfrak{F}}_M(\tau) \mid \tau' \models \neg C\}$$

(over finite-change removals; Section 4.2; under count-only $\preceq_\tau$ this is the full minimum-count set). The valid-cause set is

$$\begin{aligned}\mathsf{Causes}_{M,\mathfrak{F},\tau}(E_{\mathfrak{F}}) = \{C \in \mathcal{K}_{\mathfrak{F}} \mid\ & \tau \in \mathcal{L}(C),\ \tau \in \mathcal{L}(E_{\mathfrak{F}}),\\ & \mathrm{Close}^{\mathfrak{F}}_{\tau,M}(\neg C) \neq \emptyset,\\ & \mathrm{Close}^{\mathfrak{F}}_{\tau,M}(\neg C) \cap \mathcal{L}(E_{\mathfrak{F}}) = \emptyset\}.\end{aligned}$$

The third (non-vacuity) clause is essential: without it any C that no admissible trace violates — a tautology, or any invariant of M under an input-only frame — would pass vacuously. The fourth clause is counterfactual dependence on *every* closest removal (under count-only) or the unique closest removal (under profile refinement).

Strategy-level causes need a re-opening frame.

Proposition 1. *Let C_M be an invariant of M ($\mathcal{L}(M) \subseteq \mathcal{L}(C_M)$). Under an input-only frame, $C_M \notin \mathsf{Causes}_{M,\mathfrak{F},\tau}(E_{\mathfrak{F}})$ for every $\tau, E_{\mathfrak{F}}$.*

Proof. $\mathrm{Adm}^{\mathfrak{F}}_M(\tau) \subseteq \mathcal{L}(M) \subseteq \mathcal{L}(C_M)$, so no admissible trace violates C_M; $\mathrm{Close}^{\mathfrak{F}}_{\tau,M}(\neg C_M) = \emptyset$ and non-vacuity fails. □

Thus an invariant is *vacuous under input-only* counterfactuals; testing it requires re-opening the machine's commitments, the principled regime for which is strategy-structure intervention (the reverse parity game).

4.4 Anti-circularity

$\mathcal{K}_{\mathfrak{F}}$ must be constrained, or the test admits circular "causes" that re-assert the effect. The key distinction is between *causing E through M* (what a genuine cause does, and must be allowed) and *restating E* (asserting the effect at its own evaluation positions, which is circular). The condition bans the latter without touching the former.

Assumption 2 (Anti-circularity)**.** *Every $C \in \mathcal{K}_{\mathfrak{F}}$ is* effect-disjoint*:*

(i) ***No restatement (semantic, point/bounded effects).*** *For a point effect $E_{\mathfrak{F}} = X^k o$, C does not constrain the value of o at position k: $\mathcal{L}(C)$ is invariant under flipping o at position k (i.e. $\tau' \in \mathcal{L}(C) \iff \tau'[k{:}o{\mapsto}\neg o] \in \mathcal{L}(C)$, where the bracket flips only o at time k). Equivalently, in the masked vocabulary that hides $o@k$, C is well-defined. This is the clean, semantic form of "C does not assert the effect at its own position"; it admits a cooldown cause "o at $t = 5$" for effect "o at $t = 10$" ($t \neq k$) and any inputs, while barring $X^k o$, $X^k \neg o$, and trivial re-encodings such as $X^k(o \vee \bot)$. For a bounded conjunctive effect the masking is applied at each constrained position.* For a general ω-regular effect *($GF\,grant$, $G(req \to F\,grant)$, parity acceptance) there is no single "effect position" to mask; guard (ii) and a declared* effect cone *(Assumption 3, mandatory) apply instead.*

(ii) ***No pure-logic entailment (all effects).*** *$\mathcal{L}(C) \not\subseteq \mathcal{L}(E_{\mathfrak{F}})$, decidable by emptiness of $\mathcal{L}(C) \cap \overline{\mathcal{L}(E_{\mathfrak{F}})}$ (an ω-regular inclusion test). The test is* not *relativized to $\mathrm{Adm}_M^{\mathfrak{F}}(\tau)$: a candidate that forces $E_{\mathfrak{F}}$ only* through M *is* not *excluded (this is what reactive causes do); only one that forces $E_{\mathfrak{F}}$ by pure logic — already a logical strengthening of the effect — is. This bars logical entailment and point-position restatement; it does* not *exclude system-relative aliases or near-restatements that pass (ii) alone (hence the mandatory effect cone below).*

We call this "precise and decidable" in the sense that (i) is a decidable semantic invariance check for point/bounded effects, (ii) is a decidable ω-regular inclusion test for all effects, and the effect cone is a declared finite AP subset; we do not *claim a uniform syntactic "effect-position" notion for arbitrary temporal effects.*

Assumption 3 (Effect cone (mandatory for general effects))**.** *For every non-point effect $E_{\mathfrak{F}}$, the frame declares a finite* AP*-subset $\mathsf{Cone}(E_{\mathfrak{F}})$ of propositions that may appear in effect-shaped subformulas (outputs and acceptance atoms tied to E). Every $C \in \mathcal{K}_{\mathfrak{F}}$ must satisfy $\mathrm{AP}(C) \cap \mathsf{Cone}(E_{\mathfrak{F}}) = \emptyset$ unless C is explicitly marked, in $\mathfrak{F}_{\mathrm{strat}}$, as an internal-state observable (e.g. at_s) or a declared system-guarantee invariant. This blocks vacuous near-restatements that name the effect atom — e.g. a bare recurrence $GF\,grant$, or a guarded re-assertion of $grant$ — for effect $GF\,grant$ when* $\mathsf{Cone} = \{grant\}$*, while still allowing input causes and, under a declared regime, strategy-level state observables and system-guarantee invariants (e.g. the controller commitment $G(req \to F\,grant)$, admitted only by such an explicit declaration, never at the trace level).*

This is what makes the worked causes admissible. For the general effect $F\,grant$, guard (i) is silent and (ii) is operative: the input cause req satisfies $\mathcal{L}(req) \not\subseteq \mathcal{L}(F\,grant)$ (a req-at-0, grant-free trace witnesses this), so it is admitted — yet req does force $F\,grant$ on $\mathcal{L}(M)$, which is precisely why it is a cause. The TempoBench cause $\Gamma^\star$ (Section 9), pinning o at k through the functional machine, is admitted similarly (for the point effect $X^k o$, guard (i) masks only $o@k$, which $\Gamma^\star$, being over inputs, does not touch). The invariant $G(req \to F\,grant)$ names the cone atom $grant$, so when $\mathsf{Cone}(F\,grant) = \{grant\}$ it is admitted *only* as a declared system-guarantee invariant under $\mathfrak{F}_{\mathrm{strat}}$ (never at the trace level); given that declaration it is not a logical restatement either, since a request-free trace satisfies it vacuously while failing the effect. *Excluded* are logical restatements:

$X\,grant$ and $req \wedge X\,grant$ entail $F\,grant$ (fail (ii)); $X^k o$ and $X^k(o \vee \bot)$ for point effect $X^k o$ fail (i).

Remark 3 (Strategy-level state observables). Under $\mathfrak{F}_{\text{strat}}$, candidates may use *at_s* atoms (exposing internal state). Anti-circularity (ii) still requires $\mathcal{L}(C) \not\subseteq \mathcal{L}(E_{\mathfrak{F}})$; coincidence of C with E on $\mathcal{L}(M)$ is machine entailment, not logical restatement. Such observables are admissible when the explanatory question is about a controller *commitment*, not a trace event (Section 6.1).

4.5 Minimal cause and the preference order

Definition 2 (Minimal temporal cause)**.** *$C^\star \in \mathcal{K}_{\mathfrak{F}}$ is a minimal temporal cause iff $C^\star \in \mathsf{Causes}_{M,\mathfrak{F},\tau}(E_{\mathfrak{F}})$ (validity) and no $C' \in \mathsf{Causes}_{M,\mathfrak{F},\tau}(E_{\mathfrak{F}})$ has $C' \lhd C^\star$ (minimality). When $\lhd$ is partial, several incomparable minimal causes may exist — "*a *minimal cause," not "*the.*" We assume $\lhd$ well-founded (automatic when $\mathcal{K}_{\mathfrak{F}}$ is finite).*

Validity already encodes necessity; $\lhd$ selects *among* necessary causes. Taking permissiveness (larger $\mathcal{L}(C)$) as the *primary* key is defective: if A, B are each valid then $A \vee B$ is weaker, so a permissiveness-first order drifts to ever-larger disjunctions and would even prefer $F\,req$ to req. We therefore make *syntactic readability primary and permissiveness only a tie-break*; the canonical order is

$$(\text{smaller formula size, smaller temporal depth, more permissive (larger } \mathcal{L}(C))),$$

lexicographic. For "size" and "depth" to be reproducible we fix a candidate grammar (the readable LTL grammar of Section 5) and a normal form (negations on literals, fixed $\wedge, \vee$ ordering, no redundant X-nesting), measuring size as syntax-tree node count; the grammar and measure are part of the frame. We do not claim this priority is uniquely correct — it is a declared design choice; the frame may substitute another well-founded order. The default has three consequences: (i) it avoids overfitting (large conjunctions lose on size); (ii) it avoids disjunctive drift ($A \vee B$ is larger, chosen only when no smaller cause exists — exactly overdetermination, Section 7); (iii) the degenerate extensionally maximal object below has no small readable formula. We require of $\mathcal{K}_{\mathfrak{F}}$ that the trace-exclusion artifact $A \setminus \{f_0\}$ be inexpressible (it holds for the readable grammar/template families). If the necessary condition lies outside $\mathcal{K}_{\mathfrak{F}}$ (e.g. $a \vee b$ when only literals are admitted), the method reports $\mathsf{Causes} = \emptyset$; the remedy is to enlarge $\mathcal{K}_{\mathfrak{F}}$, not to return a spurious singleton.

Proposition 1 (Decidability for fixed finite candidate space)**.** *Fix $(M, \mathfrak{F}, \tau, E_{\mathfrak{F}})$ with $|\mathcal{K}_{\mathfrak{F}}| < \infty$. For each $C \in \mathcal{K}_{\mathfrak{F}}$, membership in $\mathsf{Causes}_{M,\mathfrak{F},\tau}(E_{\mathfrak{F}})$ is decidable (N1: non-emptiness plus shortest-edit search, Remark 1; anti-circularity is decidable by Assumption 2). If $\lhd$ is well-founded on $\mathcal{K}_{\mathfrak{F}}$, a $\lhd$-minimal valid $C^\star$ exists and is found by finite search. This is the precise "compute a minimal cause" claim; unbounded LTL grammar search is only decidable under an explicit size bound (Section 8).*

4.6 Two distinct necessity notions

We state *once* the central distinction (it recurs throughout and we cross-reference here rather than repeat it).

	N1 (trace-existential)	**N2 (strategy-guarantee)**
Inputs	$M, \mathfrak{F}, E, \tau$	$G_\varphi, \sigma, V_{\mathfrak{F}}, \mathfrak{F}, E$
Centering	actual trace τ	actual strategy σ
Question	closest $\neg C$ trace fails E?	can σ drop C and still guarantee E?
Machinery	one-player shortest-edit + non-emptiness	reverse parity game + re-imposed objective
Liveness	fails if $\mathrm{Close} = \emptyset$	primary regime

Trace-existential necessity (N1)

the literal Causes definition above: every closest admissible $\neg C$-trace (under count-only $\preceq_\tau$, or the unique one under profile refinement) must fail E. Deciding N1 for fixed C is non-emptiness of $\mathrm{Adm} \cap \mathcal{L}(\neg C)$ over finite-change removals plus a shortest-edit search in the product $M \times A_{\neg C} \times A_E$ (Remark 1), then evaluating E on the result — not emptiness alone. Token-level: "this execution."

Strategy-guarantee necessity (N2)

the two-player, strategy-level question: can the system *drop the commitment* C and still *guarantee* E against every *admissible* environment? "Admissible" means satisfying the background $B_{\mathfrak{F}}$ (e.g. a fairness assumption); quantifying over *all* environments, including $B_{\mathfrak{F}}$-violating ones, would let an unfair environment trivially falsify any liveness commitment, so both clauses are relativized to $B_{\mathfrak{F}}$. Formally, the frozen σ is assumed to secure the guarantee and to satisfy C (the actuality requirement at the strategy level); C is *strategy-necessary* iff **no** $\sigma' \in \Sigma_{\mathfrak{F}}$ both

(i) *secures* E on admissible plays: $\forall \rho.\ \mathrm{Play}(\sigma', \rho) \models (B_{\mathfrak{F}} \to E)$, and

(ii) is *not* C-committed on an admissible play: $\exists \rho.\ \mathrm{Play}(\sigma', \rho) \models B_{\mathfrak{F}} \wedge \neg C$.

Type-level: "this controller's guarantee." (When $B_{\mathfrak{F}} = \mathsf{true}$, (i) is $\forall \rho.\, E$ and (ii) is $\exists \rho.\, \neg C$, as in the un-relativized form.) N2 is centered on the actual *strategy* σ, not on τ: it is a property of the controller, so the trace-level centering of τ plays no role here (cf. Remark 2).

The reverse parity game (Section 6) decides N2; N1 is the one-player check. We do *not* claim a single parity game decides N1.

4.7 The certificate $C^\flat$

Write $A = \mathrm{Adm}^{\mathfrak{F}}_M(\tau)$ and $F = \{\tau' \in A \mid \tau' \not\models E_{\mathfrak{F}}\}$ (admissible effect-failures), and $\uparrow_{\preceq_\tau}(F) = \{\tau'' \mid \exists \tau' \in F.\ \tau' \preceq_\tau \tau''\}$ (here "above" means *farther* from τ). Let $C^\flat$ be *any* ω-regular property with $\mathcal{L}(C^\flat) \cap A = A \setminus \uparrow_{\preceq_\tau}(F)$; this pins $C^\flat$ only on A, so "$C^\flat$" names an equivalence class, not a unique formula.

Proposition 2 (Nearest-failure-anchored certificate)**.** *Suppose Assumption 1 holds, $\preceq_\tau$ is well-founded on A, $\tau \models E_{\mathfrak{F}}$, and $F \neq \emptyset$. Then:*

(a) $C^\flat$ satisfies the four valid-cause clauses*; hence it is a valid cause* provided $C^\flat \in \mathcal{K}_{\mathfrak{F}}$*. As a distance-defined object $C^\flat$ need not lie in $\mathcal{K}_{\mathfrak{F}}$ (it may be unreadable or fail effect-disjointness), so we do not assert $C^\flat \in$ Causes unconditionally; it is a witness, not the returned answer $C^\star$;*

(b) $\mathrm{Close}^{\mathfrak{F}}_{\tau,M}(\neg C^\flat) = \min_{\preceq_\tau}(F)$*;*

(c) among valid causes whose closest removals are the globally nearest effect-failures, $C^\flat$ is the smallest-language *one: $\mathcal{L}(C^\flat) \cap A \subseteq \mathcal{L}(C) \cap A$ for every such C. Equivalently — and this is not a contradiction — its admissible models $\mathcal{L}(C^\flat) \cap A = A \setminus \uparrow(F)$ form the* largest *failure-free neighborhood of τ, because its complement $\uparrow(F)$ is the* smallest *failure-containing up-set; the cause is language-minimal precisely because the failures it must admit are pushed as far out as possible.*

Proof. Within A, $\neg C^\flat \equiv\uparrow(F)$. For (b): by antisymmetry and well-foundedness $\min(\uparrow(F)) = \min(F)$ (if $m \in \min(\uparrow F)$ then some $f \in F$ has $f \preceq_\tau m$, minimality gives $m \preceq_\tau f$, and antisymmetry $m = f \in F$; the converse is similar). For (a): $\min(F) \neq \emptyset$ by the Limit Assumption (non-vacuity); every closest removal lies in F, failing $E_{\mathfrak{F}}$; $\tau \models E_{\mathfrak{F}}$ by hypothesis and $\tau \models C^\flat$ since centering gives $\tau \notin\uparrow(F)$. For (c): let C be valid with $\mathrm{Close}^{\mathfrak{F}}_{\tau,M}(\neg C) = \min_{\preceq_\tau}(F)$. If $\tau' \in \mathcal{L}(C^\flat) \cap A$ but $\tau' \notin \mathcal{L}(C)$, then $\tau' \models \neg C$, so some $\tau'' \in \mathrm{Close}^{\mathfrak{F}}_{\tau,M}(\neg C)$ satisfies $\tau'' \preceq_\tau \tau'$ (well-founded minimality). Since $\mathrm{Close}^{\mathfrak{F}}_{\tau,M}(\neg C) = \min(F) \subseteq F$, we have $\tau'' \in F$, hence $\tau' \in\uparrow(F)$, contradicting $\tau' \in A \setminus\uparrow(F) = \mathcal{L}(C^\flat) \cap A$. □

Remark 4 (The extensionally maximal cause is the wrong target). The *extensionally maximal* (largest-language) valid cause — we avoid "weakest" to prevent clash with Dijkstra's weakest precondition — is degenerate: removing a single nearest failure f_0 gives the valid cause $A \setminus \{f_0\}$, maximal yet non-explanatory and arbitrary in f_0. The *answer* is the $\triangleleft$-minimal readable $C^\star$; $C^\flat$ is a *theoretical certificate* (smallest anchored language on A, generally unreadable, not necessarily in $\mathcal{K}_{\mathfrak{F}}$); it lower-bounds $\mathcal{L}(C) \cap A$ among anchored valid causes but does not by itself yield $C^\star$ — search still enumerates $\mathcal{K}_{\mathfrak{F}}$ or runs CEGIS (Section 5). The extensionally maximal artifact is excluded (unreadable and large). These three objects — $C^\star$ (answer), $C^\flat$ (certificate anchor), extensionally maximal (excluded) — must not be conflated. $C^\flat$ stays ω-regular when $\uparrow_{\preceq_\tau}(F)$ is (true for the count-first order via Remark 1 on lassos; otherwise an explicit automaticity hypothesis). We do not claim the order *relation* is automatic in general.

A useful output is the triple $(C^\star, W, \Pi)$: the cause $C^\star$; the preserved region W (accepting region of $P = M \times A_E$ in one-player regimes, or $W_\exists$ in strategy-structure); and counterfactual witnesses Π (one per rejected competitor C', each a lasso $\pi_{C'} \in \mathrm{Close}^{\mathfrak{F}}_{\tau,M}(\neg C') \cap \mathcal{L}(E_{\mathfrak{F}})$ from the emptiness/shortest-edit check or Proposition 3).

4.8 Canonical frames

To answer the objection that a free choice of $\mathfrak{F}$ injects the cause, we fix canonical defaults; results are reported against one unless a deviation is justified.

- $\mathfrak{F}_{\text{trace}}$ — $B_{\mathfrak{F}} = \mathsf{true}$ (or stated environment assumptions), input-only, count-only $\preceq_\tau$ (Definition 1: no profile tie-break unless declared), effect-disjoint candidates over $I \cup O$ with declared $\mathsf{Cone}(E_{\mathfrak{F}})$, and $\triangleleft =$ (size, depth, permissiveness). Default for one execution; the toy example uses optional profile refinement for a unique witness.
- $\mathfrak{F}_{\text{strat}}$ — as above but strategy-structure over stated $(G_\varphi, \sigma, V_{\mathfrak{F}})$ with edited-vertex order refined by trace distance (Remark 2), internal-state observables allowed (Remark 3), tested via Section 6.
- $\mathfrak{F}_{\text{TB}}$ — the finite-horizon, point-effect, input-only frame of Section 9.

Two safeguards: every component but candidate vocabulary, grammar, similarity tie-break, and effect cone is fixed by the regime; and under a *fully specified* canonical frame (including those declarations) $\mathsf{Causes}_{M,\mathfrak{F},\tau}(E_{\mathfrak{F}})$ is a function of $(M, \tau, E_{\mathfrak{F}})$ alone. Frame-relativity means results are conditional on that declaration, as background conditions are in ordinary causal explanation. Determinacy requires stating vocabulary, grammar, count-only vs. profile $\preceq_\tau$, and $\mathsf{Cone}(E_{\mathfrak{F}})$ for non-point effects.

5 Trace-Level Construction and Toy Example

5.1 The trace-level procedure (one-player)

Given $(M, \mathfrak{F}, E, \tau)$, the trace-level test (N1) is: (1) compile E into a deterministic parity automaton $A_E = (Q, q_0, 2^{I \cup O}, \delta_E, \Omega)$ and form the frozen product $P = M \times A_E$ (a *one-player* arena: M is fixed, so only the environment moves); (2) for a candidate C, build $A_{\neg C}$ and check

$$\mathrm{Close}^{\mathfrak{F}}_{\tau,M}(\neg C) \neq \emptyset \quad \text{and} \quad \mathrm{Close}^{\mathfrak{F}}_{\tau,M}(\neg C) \cap \mathcal{L}(E) = \emptyset,$$

an emptiness check plus shortest-edit search (Remark 1). To *search* for a $\lhd$-minimal valid cause when $|\mathcal{K}_{\mathfrak{F}}| < \infty$, enumerate candidates (Theorem 1); otherwise use bounded grammar/template search with CEGIS:

$$C ::= p \mid \neg C \mid C \wedge C \mid XC \mid FC \mid GC \mid C\,U\,C,$$

or templates $G(a \to Fb), F(a \wedge Xb), G(a \to Xb), GFa, FGa$, proposing C, checking N1, and refining from witnesses Π. The certificate $C^\flat$ (Section 4.7) characterizes the anchored language lower bound but does not replace this search. Strategy-level minimality uses dominion subset search (Section 6). The returned formula need not be a subformula of φ; we translate readable LTL from $\mathcal{K}_{\mathfrak{F}}$ when needed.

5.2 Toy example

Let $\varphi = G(req \to F\,grant)$ and let M have states $idle, pending$ with initial state $s_0 = idle$, specified completely:

$$\begin{array}{rcll} \delta(idle, req) & = & pending, & \qquad \lambda(idle, \cdot) = \emptyset, \\ \delta(idle, \neg req) & = & idle, & \qquad \lambda(pending, \cdot) = \{grant\}, \\ \delta(pending, \cdot) & = & idle. & \end{array}$$

A request in *idle* moves to *pending* next; in *pending* the machine emits *grant* and returns to *idle*; a request in *pending* is *not* latched. The machine is effectively Moore (*grant* holds exactly in *pending*). The actual trace is $\{req\}, \{grant\}, \emptyset, \emptyset, \ldots$ and the effect is $E = F\,grant$.

The minimal trace-level cause is *req*. Use $\mathfrak{F}_{\text{trace}}$ with count-only $\preceq_\tau$ ($\mathrm{Adm} = \mathcal{L}(M)$). Every minimum-count removal sets $\neg req$ at time 0: the machine stays in *idle*, never enters *pending*, never grants; $d = 2$ and $F\,grant$ fails. No $\neg req$-removal of count ≤ 2 preserves the effect (emitting *grant* needs a *req* in *idle* at some $j \geq 1$; the cheapest such trace has $d = 3$). All minimum-count removals fail $F\,grant$; so *req* is non-vacuous and necessary — a valid cause. (Under profile refinement the unique closest removal is the $d = 2$ trace above; under Baire or inclusion orders *req* would spuriously fail.) The candidate $req \wedge X\,grant$ is barred by Assumption 2(ii) and loses on size. Hence $C_\tau = req$.

Why *req* can feel uninformative. That "a request alone does not explain *why* the machine granted" is a critique of the input-only *frame*, which fixes the machine's response rule as background: relative to that, the only thing that had to hold for the grant was that a request arrived. To make the response rule a candidate, move to a strategy frame. The strategy-level target is $C_M = G(req \to F\,grant)$, or, exposing the state, $G((at\_idle \wedge req) \to X\,pending) \wedge G(pending \to grant)$ (the first conjunct must be guarded by *at_idle*, since a request in *pending* is not latched, so unguarded $G(req \to X\,pending)$ is not an invariant). These entail $G((at\_idle \wedge req) \to X\,grant)$, hence $G(req \to F\,grant)$ on traces whose requests arrive only in *idle*. By Proposition 1, C_M is testable only under a strategy frame; it names the cone atom *grant*, so when $\mathsf{Cone}(F\,grant) = \{grant\}$ it is admissible only because $\mathfrak{F}_{\text{strat}}$ declares it a system-guarantee invariant (Assumption 3), not at the trace level. The two answers — *req* (trace) and $G(req \rightarrow F\, grant)$ (strategy) — are outputs of two different frames, not competitors.

6 The Reverse Parity Game (Strategy Level)

Many temporal effects ($GF\, a$, $FG\, stable$, $G(req \rightarrow F\, ack)$) cannot be explained by a single finite event; the parity condition is needed to reason about recurrence. But it is needed *only* for N2: in the input-only and output-intervention regimes the trace test is one-player (Section 4.6). This section gives the genuine two-player object for N2. ("Reverse" here names the *explanatory* direction — we run the synthesis machinery backwards, from a frozen controller to the conditions it needed — not a retrograde or backward-time analysis; the game itself is a standard forward parity game on a partial-commitment arena.) Fix the synthesis game $G_\varphi = (V, V_0, V_1, T, \Omega)$, the frozen σ implemented by M, the candidate C with deterministic parity monitor $A_{\neg C}$, the effect monitor A_E, and the editable vertex set $V_{\mathfrak{F}} \subseteq V_0$ defining $\Sigma_{\mathfrak{F}}$.

Definition 3 (Reverse parity game)**.** *$G^{\text{rev}}_{M,\mathfrak{F},C,E} = (V^r, V^r_0, V^r_1, T^r, \Omega^r)$ has vertices $V^r = V \times Q_B \times Q_E \times Q_{\neg C}$, pairing a game vertex with the states of deterministic monitors A_B (for the background $B_{\mathfrak{F}}$), A_E, and $A_{\neg C}$. It is* partitioned *so that V^r_0 ($\exists$, the re-opened system) are vertices over $V_{\mathfrak{F}}$, where $\exists$ may take any G_φ-move, and V^r_1 ($\forall$) are all others: vertices over V_1 (environment) and over the frozen $V_0 \setminus V_{\mathfrak{F}}$ (single-successor, assigned to $\forall$ to keep a genuine partition while leaving play unaffected). Edges carry the synthesis-game label $\ell : T \rightarrow 2^{I \cup O}$; T^r updates V by T and each monitor by its transition function on ℓ. The $\exists$-objective is the assume-guarantee condition $B_{\mathfrak{F}} \rightarrow E$: $\exists$ wins π iff $\pi|_{I \cup O} \models B_{\mathfrak{F}} \rightarrow E$. This is ω-regular, so Ω^r is a parity index for it and G^{rev} is a genuine parity game; A_B and $A_{\neg C}$ are also read off as* auxiliary *acceptance conditions used in the witness search below.*

N2 is a *conjunctive* strategy-existence question — $\exists \sigma'$ that secures $B_{\mathfrak{F}} \rightarrow E$ ($\forall \rho$) *and* admits an admissible $\neg C$ play ($\exists \rho.\ B_{\mathfrak{F}} \wedge \neg C$) — and ordinary parity solving returns one winning strategy and the winning region, not the set of all winning strategies. **Caution:** for a parity/Büchi objective, staying inside the winning region forever does *not* imply winning (a play may remain in $W_\exists$ yet violate the parity condition), so we do *not* use a "trap" multi-strategy of "all moves staying in $W_\exists$"; we re-impose the objective explicitly in the witness search.

Proposition 3 (Reverse-game soundness for N2)**.** *Assume the frozen $\sigma \in \Sigma_{\mathfrak{F}}$ secures $B_{\mathfrak{F}} \rightarrow E$ and satisfies C (the strategy-level actuality requirement). Let $W_\exists$ be the $\exists$-winning region of $G^{\text{rev}}_{M,\mathfrak{F},C,E}$ for objective $B_{\mathfrak{F}} \rightarrow E$, and let $\mathcal{A}[W_\exists]$ be the sub-arena obtained by restricting G^{rev} to $W_\exists$ (keeping only edges that stay in $W_\exists$). Then C is* not *strategy-necessary (N2 fails) iff $\mathcal{A}[W_\exists]$ contains a play π with*

$$\pi|_{I \cup O} \models B_{\mathfrak{F}} \wedge E \wedge \neg C.$$

Equivalently, C is strategy-necessary iff the product $\mathcal{A}[W_\exists] \times A_B \times A_E \times A_{\neg C}$ has no accepting lasso for the conjunction $B_{\mathfrak{F}} \wedge E \wedge \neg C$ — a one-player ω-regular non-emptiness check (not mere reachability of a vertex).

Proof. Let Π_{win} be the plays consistent with *some* $\sigma' \in \Sigma_{\mathfrak{F}}$ that secures $B_{\mathfrak{F}} \rightarrow E$. We claim $\Pi_{\text{win}} = \{\pi : \pi \text{ stays in } W_\exists \text{ and } \pi \models B_{\mathfrak{F}} \rightarrow E\}$. ($\subseteq$) A securing strategy keeps every play in $W_\exists$ and wins it, so its plays satisfy $B_{\mathfrak{F}} \rightarrow E$. ($\supseteq$) Given such a π, we may take $\pi = uv^\omega$ to be an accepting *lasso*: $\mathcal{A}[W_\exists]$ is finite and the winning condition ω-regular, so whenever a qualifying play exists a lasso one does (this is the lasso/non-emptiness fact, standard for deterministic ω-automata over a finite arena). Define σ' to follow the (ultimately periodic) moves of π while the history matches π, and a fixed *finite-memory* winning strategy from $W_\exists$ after any environment deviation: every $\exists$-move on π goes to $W_\exists$ (π stays there), and the environment cannot escape $W_\exists$. Following a lasso needs only finite memory (the period), and finite-memory winning strategies exist for parity objectives on finite arenas, so σ' is finite-memory; hence $\sigma' \in \Sigma_{\mathfrak{F}}$ (which is the finite-memory class by default, Section 4), σ' secures $B_{\mathfrak{F}} \to E$, and π is one of its plays. Now C is not strategy-necessary iff some σ' secures $B_{\mathfrak{F}} \to E$ (i) and admits a play with $B_{\mathfrak{F}} \wedge \neg C$ (ii), i.e. iff Π_{win} contains a play with $B_{\mathfrak{F}} \wedge \neg C$. By the claim such a play stays in $W_\exists$ and satisfies $B_{\mathfrak{F}} \to E$; with $B_{\mathfrak{F}}$ this gives E, so it satisfies $B_{\mathfrak{F}} \wedge E \wedge \neg C$ and lies in $\mathcal{A}[W_\exists]$. Conversely any $\mathcal{A}[W_\exists]$-play with $B_{\mathfrak{F}} \wedge E \wedge \neg C$ satisfies $B_{\mathfrak{F}} \to E$ and stays in $W_\exists$, hence lies in Π_{win} and witnesses (i)+(ii). The lasso/non-emptiness characterization is standard for deterministic ω-automata over a finite arena. $\square$

This decision is sound *and* complete and uses only (a) a two-player parity solve for the winning region $W_\exists$ and (b) a one-player ω-regular non-emptiness check on $\mathcal{A}[W_\exists]$. It does *not* rely on a maximal-permissive-strategy construction, sidestepping the fact that for parity objectives the "moves staying in $W_\exists$" multi-strategy is *not* winning (a Bernet–Janin–Walukiewicz most-permissive *winning* strategy needs memory); the crucial point is that the objective $B_{\mathfrak{F}} \to E$ is re-imposed in the witness search, so non-winning plays that merely stay in $W_\exists$ are excluded. The input-only/output regimes are the degenerate case $V_{\mathfrak{F}} = \emptyset$ ($V_0^r = \emptyset$): the game is one-player and the question collapses to the trace check of Section 5 (N1). Minimizing $V_{\mathfrak{F}}$ is a separate optimization (Section 8).

From commitments to a trace cause. If C is strategy-necessary (no $\mathcal{A}[W_\exists]$-play satisfies $B_{\mathfrak{F}} \wedge E \wedge \neg C$), the witness is the set $D_C \subseteq V_{\mathfrak{F}}$ of editable vertices whose σ-move, if re-opened, would let $\exists$ secure $B_{\mathfrak{F}} \to E$ via a $\neg C$-deviation, with transitions $T_C = \{(v, \sigma(v)) \mid v \in D_C\}$. For a *safety* commitment the induced cause is $C \equiv G \bigwedge_{v \in D_C}(at\_v \to move_\sigma(v))$ (negation $F \bigvee(\ldots)$); for a *liveness* commitment, e.g. $C = GF\, at\_serve$ below, it is $GF \bigvee_{v \in D_C} at\_v$. Breaking T_C is exactly what a $\neg C$-deviation does.

Minimal strategy cause. The strategy level so far has only *validity* (N2). A *minimal strategy cause* is a $\subseteq$-minimal editable set $V_{\mathfrak{F}}$ (equivalently a minimal dominion D_C) under which C is strategy-necessary. Unlike the trace level, this minimization is not a single parity solve but a subset search with the parity decision as oracle (Section 8); we state plainly that the strategy level delivers validity via the game and minimality only via this separate, harder optimization.

6.1 Worked liveness example

The toy effect $F\, grant$ is reachability (co-safety) and needs no recurrence. For a parity effect, take a two-state machine over input req, output $grant$:

$$\delta(serve, \cdot) = wait,\ \ \delta(wait, req) = serve,\ \ \delta(wait, \neg req) = wait; \quad \lambda(serve, \cdot) = \{grant\},\ \lambda(wait, \cdot) = \emptyset.$$

Being Moore, the machine emits *grant* exactly *while* in *serve* (not the step after), entering *serve* one step after a request in *wait*. The initial state is $s_0 = wait$. Effect $E = GF\, grant$; actual trace $\tau = (\{req\}, \{grant\})^\omega$ (i.e. $a_0 = \{req\}$, $a_1 = \{grant\}$, ...), which grants infinitely often.

Fairness is required. No controller guarantees $GF\, grant$ against *every* environment: one issuing finitely many requests traps the machine in *wait*. The well-posed question carries the environment-fairness background $B_{\mathfrak{F}} = GF\, req$, and the guaranteed objective is $GF\, req \to GF\, grant$, again

ω-regular. Without $B_{\mathfrak{F}}$ the winning region is empty and every candidate is vacuously "necessary," so stating $B_{\mathfrak{F}}$ is what makes the example meaningful.

The monitor A_E for $GF\,grant$ is a two-priority parity (Büchi) automaton (priority 0 on *grant*, 1 elsewhere; accepting iff 0 recurs); a single priority cannot encode "infinitely often." The product is accepting iff the play visits *serve* infinitely often — genuine recurrence, not reachability.

The strategy-level cause is the recurrence commitment. Take $C = GF\,at\_serve$ (effect-disjoint: it names *at_serve*, and $\mathcal{L}(GF\,at\_serve) \not\subseteq \mathcal{L}(GF\,grant)$ by pure logic, coinciding only through M). Under $\mathfrak{F}_{\text{strat}}$ with $V_{\mathfrak{F}}$ the *wait*-vertices and $B_{\mathfrak{F}} = GF\,req$, apply Proposition 3. The witness search asks for a play in $\mathcal{A}[W_\exists]$ satisfying $GF\,req \wedge GF\,grant \wedge \neg C$, i.e. $GF\,req \wedge GF\,grant \wedge FG\,at\_wait$. But *grant* is emitted only in *serve*, so $GF\,grant$ entails $GF\,at\_serve$, which contradicts $FG\,at\_wait$; no such play exists, so C is strategy-necessary. (Note this uses the $B_{\mathfrak{F}}$-relativized N2: an unfair environment with finitely many requests would, against the frozen σ, yield a play with $FG\,at\_wait = \neg C$, but that play violates $B_{\mathfrak{F}} = GF\,req$ and is therefore not an admissible deviation witness; this is exactly why clause (ii) is relativized to $B_{\mathfrak{F}}$.) The dominion D_C is the *wait*-vertices from which σ serves on a request — the accepting-recurrence-preserving commitment. By contrast the *stronger* "serve every request" $G((at\_wait \wedge req) \to X\,at\_serve)$ is *not* necessary (a deviation may skip some requests yet serve infinitely many, securing the objective on a $B_{\mathfrak{F}}$-fair play while violating the invariant), and the reachability candidate "*grant* at time 1" is likewise not necessary. Only a recurrence-preserving commitment passes — why parity, not reachability, is the right acceptance here.

7 Overdetermination, Preemption, and Necessity vs. Sufficiency

Pure necessity has known limitations under overdetermination and preemption; we address them head-on.

Overdetermination. If inputs a and b each force $E = X\,grant$ and both hold, neither singleton is necessary (removing a leaves b, which still forces the grant), so neither is valid — the correct report under necessity. The *necessary condition* (not necessarily "the actual cause" in a stronger sense) is the disjunction $C = a \vee b$, whose closest removal sets both false and destroys E — *provided* $\mathcal{K}_{\mathfrak{F}}$ admits disjunctions. This does not conflict with the readability-first preference order (Section 4), which penalizes disjunctions on size: $a \vee b$ is dispreferred to any smaller valid singleton, so it is returned *only* when no smaller valid cause exists — which is exactly the overdetermined case, where there genuinely is no smaller necessary cause. If $\mathcal{K}_{\mathfrak{F}}$ is restricted (e.g. to literals), no singleton is valid and the method reports $\mathsf{Causes} = \emptyset$; the remedy is to enlarge $\mathcal{K}_{\mathfrak{F}}$. In this symmetric example $C^\flat$ coincides with $a \vee b$ on the admissible set, but this identification is special to the example: in general $C^\flat$ is the distance-defined neighborhood and need not equal any readable disjunction.

Preemption. If a forces E and b would have only had a been absent, then on a trace with both, a alone is not necessary; the necessary condition is again $a \vee b$. Distinguishing preempting from preempted needs sufficiency/contingency, which necessity does not provide. A frame wanting this imports a Halpern–Pearl-style contingency by pinning b in $B_{\mathfrak{F}}$, recovering a relative to that frame.

Why necessity. Two objections. "Necessity flags trivial conditions ('the system was powered on')." It does not: non-vacuity discards conditions no admissible counterfactual can violate, and $B_{\mathfrak{F}}$ absorbs standing operating assumptions into the setting. "Why not sufficiency (Halpern–Pearl)?"

HP adds a minimal-sufficient-set requirement with a contingency quantifier; even in the Boolean setting its complexity is definition-sensitive — NP-complete for the original definition in binary models, Σ_2^P-complete for the original definition in general models, D_2^P-complete for the updated definition (Eiter–Lukasiewicz; Aleksandrowicz–Chockler–Halpern–Ivrii). Lifting any of these to ω-regular effects layers the contingency quantifier atop language operations, so HP sufficiency is at least as expensive as — and conceptually heavier than — our necessity test (a one-player shortest-edit plus non-emptiness check, or, at the strategy level, a parity solve followed by a non-emptiness check). We claim only that necessity is *no harder* (and uniform and decidable), not that sufficiency is *strictly* harder over ω-regular effects (which would require fixing an HP-over-ω-regular semantics we do not develop). We thus target necessity by design, report overdetermined effects as disjunctive causes when expressible, and expose contingency through $B_{\mathfrak{F}}$.

8 Decidability and Complexity

We give conditional bounds, not a fully proved complexity theorem; hypotheses are explicit. Let $|M|, |A_E|, |A_{\neg C}|$ be the machine/monitor sizes and d the priority count.

One-player (input-only / output-intervention). Candidate checking is non-vacuity (non-emptiness of $\mathrm{Adm}_M^{\mathfrak{F}}(\tau) \cap \mathcal{L}(\neg C)$ over finite-change removals) plus effect-failure. We do *not* assume the closeness *relation* is automatic (its primary key compares unbounded counts). Instead, by Remark 1, the closest removal is a shortest-edit path in the finite product $M \times A_{\neg C} \times A_E$ (size $O(|M|\,|A_{\neg C}|\,|A_E|)$), computable by BFS, then E is evaluated on it — polynomial in the product (plus parity solving $|V|^{O(d)}$, quasi-polynomial, if E is a genuine parity effect). For LTL candidates the *deterministic* parity monitors used in the explicit product are worst-case *doubly* exponential in $|C|$ (LTL $\rightarrow$ deterministic parity is doubly exponential), so the explicit shortest-edit product route is doubly exponential in the formula size. The underlying language checks (non-emptiness of $\mathrm{Adm} \cap \mathcal{L}(\neg C)$ and effect-failure) are themselves no harder than LTL model checking — PSPACE in the formula size — when carried out via *nondeterministic* monitors rather than explicit determinization; we do *not* claim a uniform PSPACE bound for the full minimum-count validity quantification. $C^\flat$ is ω-regular and computable when $\uparrow_{\preceq_\tau}(F)$ is (via the shortest-edit construction on lassos; else an explicit automaticity hypothesis).

Minimal-cause search. Over a finite template family, $|\mathcal{K}_{\mathfrak{F}}|$ checks; over the open LTL grammar, bounded-size search is decidable but exponential in the bound (as in syntax-guided synthesis).

Two-player (strategy-structure). Deciding N2 for a *fixed* $V_{\mathfrak{F}}$ (Proposition 3) is: solve the parity game for objective $B_{\mathfrak{F}} \rightarrow E$ (size $O(|V|\,|A_B|\,|A_E|)$, $O(d)$ priorities) to get the winning region $W_\exists$, then a one-player ω-regular non-emptiness check for $B_{\mathfrak{F}} \wedge E \wedge \neg C$ on $\mathcal{A}[W_\exists]$ (product with $A_B, A_E, A_{\neg C}$; an accepting-lasso search). Parity solving is in UP $\cap$ co-UP and quasi-polynomial, and the non-emptiness check is polynomial in the product, so the decision is quasi-polynomial. (We do *not* invoke a permissive-strategy computation; the winning region plus the re-imposed objective suffices.) Finding a $\subseteq$-minimal $V_{\mathfrak{F}}$ (the minimal dominion) is a *separate* search over $\leq 2^{|V_0|}$ vertex subsets, each tested by the above. We do not prove a tight class, but record an upper bound: checking that a given $V_{\mathfrak{F}}$ is necessity-inducing is one quasi-polynomial parity decision, and checking minimality is a co-NP-style universal quantification over its proper subsets (each a parity decision); so the minimal-dominion decision sits within a polynomial-hierarchy level relative to a parity oracle (e.g. a Σ_2-type search), naturally encoded as SAT/SMT/MaxSAT over vertex-selection variables with the parity decision as oracle. Exact completeness is left open.

Finite-horizon (TempoBench). $\mathcal{K}_{\mathfrak{F}_{\mathrm{TB}}}$ is finite but *exponential* ($\leq 2|I|(k+1)$ literals, hence $\leq 2^{2|I|(k+1)}$ constraint sets). Checking one candidate is a finite reachability check over A_{HOA} on the window, polynomial in $|A_{\mathrm{HOA}}|$ and k. Validity is not monotone in Γ (Remark 6), so greedy deletion is unsound; a subset-minimal cause needs exhaustive/hitting-set/CEGIS search. Each check is polynomial; the search is the hard part.

9 TempoBench Specialization

We specialize the framework to a TempoBench-style temporal causal evaluation (TCE) task. *Conditionally on the assumptions (F1)–(F2) below, which we have not verified against the released artifacts*, the solver receives the same information and solves the same finite-horizon problem, returning a *necessity*-based answer. Beyond (F1)–(F2) there is a second gap: TempoBench labels TCE with CORP, a Halpern–Pearl *actual-causation* tool in Halpern's *modified* variant (its contingency mechanism resets outputs to their actual-trace values; verified against the primary sources in the companion verification suite). The TempoBench authors confirm the shipped keys are generated with contingencies *disabled*, so each key is the contingency-free counterfactual—the subset-minimal joint input-flip—whereas we compute counterfactual *necessity*, and the two differ on overdetermined effects (Section 7: necessity returns no pivotal cell where the key is the joint flip-set). So this is an *alternate*, necessity-based account, not a reproduction of CORP's key; the actual-cause object matching the keys is developed in the companion TempoBench paper. TempoBench has two tasks: temporal trace evaluation (TTE), asking whether a finite trace is accepted by a supplied automaton, and TCE, asking for the minimal input constraints necessary for a specified output at a specified timestep.[1]

What the solver is given. An automaton A_{HOA} in HOA format, the AP list, a finite trace $\tau_{\leq n} = \alpha_0 \cdots \alpha_n$ ($\alpha_t \subseteq I \cup O$), and one or more point effects written e.g. `XXX` $g = X^3 g$. The reverse question $M + \varphi + \mathfrak{F} + E + \tau \to C^\star$ specializes to $A_{\mathrm{HOA}} + \mathfrak{F}_{\mathrm{TB}} + \tau_{\leq n} + (E_{\mathfrak{F}} = X^k o) \to C^\star$: *which provided input facts were necessary for output o at timestep k?* The effect class is $\mathcal{E}_{TB} = \{X^k o \mid o \in O_{\mathfrak{F}},\ 0 \leq k < |\tau_{\leq n}|,\ o \in \alpha_k\}$ (positive selected-output occurrences at fixed timesteps); broad effects ($F\,grant$, $G(req \to F\,grant)$, $GF\,ready$, $\mathrm{Acc}_{\mathrm{parity}}$, internal-state reachability) are extensions beyond TCE.

Remark 5 (What A_{HOA} is; faithfulness assumptions)*.* TempoBench presents the system as a finite-state *acceptor* $A_{\mathrm{HOA}} = (Q, 2^{I \cup O}, \delta, q_0, F)$: it reads valuation sequences (inputs and outputs together) and accepts a finite trace when its run from q_0 ends in F (TempoBench's TTE semantics). It is an acceptor over $I \cup O$, not an I/O transducer; $\mathcal{L}(A_{\mathrm{HOA}})$ plays the role of $\mathcal{L}(M)$, not of the effect monitor. For faithfulness we assume, *conditionally*: **(F1)** $\mathcal{L}(A_{\mathrm{HOA}})$ is exactly the system's behavior set (not a specification monitor A_φ); and **(F2)** the behavior set is *functional in the inputs* (input-deterministic), since the automata come from reactive synthesis — consistent with the acceptor framing, and the property the operational reading uses to speak of "the" output at each timestep. **We have not independently verified (F1)–(F2) against the released HOA artifacts**; whether they are input-deterministic system models rather than specification monitors is a checkable benchmark property. Accordingly the alignment is stated *conditionally on (F1)–(F2)*:

[1] TempoBench: `https://arxiv.org/abs/2510.27544`; implementation `https://github.com/nik-hz/tempobench`.

if they hold, the reduction below is exact; if the supplied automata are specification monitors, (F1) fails and the alignment does not apply. Given (F1)–(F2), the point effect $X^k o$ needs no separate monitor (it is the membership check $o \in \alpha_k$); there is no product to build, and the solver uses A_{HOA} directly.

The specialized frame. $\mathfrak{F}_{\mathrm{TB}}$ is input-only (A_{HOA} fixed; outputs, transitions, priorities not intervened on), with candidates the timestep-indexed input literals over the window $0, \dots, k$: constraint sets $\Gamma \subseteq \{0, \dots, k\} \times \mathrm{Lit}(I)$ ($\mathrm{Lit}(I) = \{i, \neg i\}$), denoting $C_\Gamma = \bigwedge_{(t,\ell)\in\Gamma} X^t \ell$. There are $\leq 2|I|(k+1)$ literals, hence $\leq 2^{2|I|(k+1)}$ candidate sets: the space is finite but *exponential*, not linear. The closeness order is the count-only coarsening, counted in changed *input cells*: under input-only intervention with (F2), outputs are determined by inputs, so a counterfactual's chosen changes are its input-cell changes, and we count those rather than the full-valuation positions of the general order (Section 4.2). So Close is the full set of minimum-count removals; minimality of causes is by set inclusion over Γ.

Finite-horizon definitions. Entirely in finite-trace terms (no infinitary acceptance for A_{HOA} is used). A finite sequence $\beta = \beta_0 \cdots \beta_n$ is *accepted* iff its run from q_0 ends in F (under (F2) the run is unique given the inputs); write $\mathcal{L}_{\leq n}(A_{\mathrm{HOA}})$ for accepted length-(n+1) sequences and $\mathrm{Adm}^{\leq k}_{A,\mathfrak{F}_{\mathrm{TB}}} = \mathcal{L}_{\leq n}(A_{\mathrm{HOA}})$. For a candidate Γ, a removal is an admissible β violating C_Γ, and

$$\mathrm{Close}^{\leq k}_{\tau_{\leq n},A}(\neg C_\Gamma) = \min_{\#\text{changes}}\{\beta \in \mathrm{Adm}^{\leq k}_{A,\mathfrak{F}_{\mathrm{TB}}} \mid \beta \not\models C_\Gamma\}.$$

$$\begin{aligned}\mathsf{Causes}^{\leq n}_{A,\mathfrak{F}_{\mathrm{TB}},k}(X^k o) = \{\Gamma \mid \tau_{\leq n} \models C_\Gamma,\ o \in \alpha_k,\ &\mathrm{Close}^{\leq k}_{\tau_{\leq n},A}(\neg C_\Gamma) \neq \emptyset,\\ &\forall \beta \in \mathrm{Close}^{\leq k}_{\tau_{\leq n},A}(\neg C_\Gamma).\ o \notin \beta_k\}.\end{aligned}$$

A minimal cause $C^\star = C_{\Gamma^\star_E}$ uses any subset-minimal valid $\Gamma^\star_E$ (several may exist).

Remark 6 (Validity is not monotone; greedy deletion is unsound). Membership is not monotone under $\subseteq$ on Γ: enlarging Γ can make a candidate valid (the cheapest removal must flip more) or invalid (the closest removal flips an irrelevant added literal while keeping the output). So "delete literals while valid" need not reach a subset-minimal cause; computing $\Gamma^\star_E$ needs genuine search (exhaustive, hitting-set/MaxSAT, or CEGIS). Each validity check is polynomial; the search is hard. Greedy becomes sound only under an extra monotonicity hypothesis, which we do not assume.

Proposition 4 (Finite specialization — a definitional correspondence)**.** *This is a* definitional *restatement, not a deep theorem: the finite-horizon objects above were* defined *to mirror the general clauses, and the content is that the mirroring is exact and needs no infinitary acceptance. We state it as a proposition only to record that correspondence precisely. With $\mathfrak{F}_{\mathrm{TB}}$ as the input-only frame, $B_{\mathfrak{F}} = \mathsf{true}$, admissible set $\mathcal{L}_{\leq n}(A_{\mathrm{HOA}})$, the change-counting order on the window, subset minimality, and (F1)–(F2), and interpreting the general* Causes *clauses over finite traces with* LTL_f *semantics, for every Γ: $\Gamma \in \mathsf{Causes}^{\leq n}_{A,\mathfrak{F}_{\mathrm{TB}},k}(X^k o)$ iff C_Γ satisfies the four valid-cause clauses on $\tau_{\leq n}$ under $\mathfrak{F}_{\mathrm{TB}}$.*

Proof. Each of the four clauses refers only to positions $0, \dots, k$ (actuality is $\tau_{\leq n} \models C_\Gamma$ and $o \in \alpha_k$; non-vacuity is non-emptiness of admissible $\neg C_\Gamma$-removals; effect-failure asks $o \notin \beta_k$ over closest removals), each determined by the length-(k+1) prefix under LTL_f. These are verbatim the clauses of $\mathsf{Causes}^{\leq n}_{A,\mathfrak{F}_{\mathrm{TB}},k}$: admissible sets coincide, the closeness order is in both cases the count-only coarsening of $\preceq_\tau$ on the window (so Close is the full minimum-count set and effect-failure quantifies over all of it), and (F2) makes "o at k" a well-defined function of the admissible inputs. No infinite object appears, so no ω-acceptance is needed. □

Evaluation compatibility. TempoBench scores at the AP level (partial credit per atomic proposition at a timestep) and the TS level (a timestep counts only if its whole constraint set matches). The output should therefore preserve timestep structure: a flat formula $X\, in_2 \wedge X^3 \neg in_0$ should also be reported grouped $(1\colon in_2,\ 3\colon \neg in_0)$. The pipeline should preserve TempoBench's difficulty features (effect depth k, automaton states, transition count, trace length, causal-input count, unique inputs) and add no hidden information from the original specification.

10 Limitations and Conclusion

Established. A frame-relative, counterfactual-necessity notion of temporal cause, decided in the common regimes by automaton emptiness/shortest-edit search and in the strategy regime by the reverse parity game with a sound-and-complete decision (Proposition 3); Proposition 2 characterizes the nearest-failure certificate; and Proposition 4 shows the finite construction is the literal specialization of the general definition, aligning it with TempoBench TCE.

Scoped / conjectural. Complexity statements (Section 8) are at the level of construction sizes and standard solving bounds, not a fully proved complexity theorem; they rely on automaticity of $\preceq_\tau$, which we establish for the canonical order on lassos and otherwise assume. The reverse game decides *strategy-guarantee* necessity (N2), a distinct type-level notion from trace-existential N1; we do not claim one game decides N1, nor that one game performs the edited-vertex minimization. The HP comparison is a contrast, not a proved separation. The TempoBench alignment is conditional on (F1)–(F2) (Remark 5), which we have not verified against the artifacts.

Limitations. The method reports frame-relative necessary conditions, not full actual causes: disjunctions for overdetermined effects, no preempting/preempted discrimination, and $\mathsf{Causes} = \emptyset$ when the necessary cause lies outside $\mathcal{K}_{\mathfrak{F}}$. Outputs are determinate only after the eligible vocabulary is declared. Evidence here is two worked examples plus the TempoBench alignment; a non-toy case study — a strategy-level extraction on a realistic arbiter, and an end-to-end TempoBench TCE instance with $\Gamma^\star$ compared to ground truth — is left to future work and is the main thing needed before empirical claims.

Conclusion. Frame-relative counterfactual necessity over a frozen synthesized strategy, with a parity-game lift for strategy-level causes, is a useful organizing frame for temporal causal explanation. This document fixes the definitions, the two regimes, and their soundness relationships; closing the remaining gaps (a proved complexity theorem, the dominion-minimization procedure, and an empirical evaluation) would turn it into a complete method.

A Compact Notation

Standard logical, temporal, set, and automaton notation is used as usual. We list only the document-specific or overloaded objects.

$\mathfrak{F}$	the causal frame $(B_{\mathfrak{F}}, E_{\mathfrak{F}}, S_{\mathfrak{F}}, O_{\mathfrak{F}}, \mathcal{K}_{\mathfrak{F}}, \mathcal{I}_{\mathfrak{F}}, \preceq, \lhd)$; explanatory scope, not a Kripke frame and not φ.
$E_{\mathfrak{F}}$	the frame-selected effect ($= E$ if no scoping); a single ω-regular trace property.
$\mathcal{K}_{\mathfrak{F}}$	candidate-cause space (effect-disjoint, Assumption 2).

$\mathcal{I}_{\mathfrak{F}}$ — intervention model fixing $\mathrm{Adm}^{\mathfrak{F}}_M(\tau)$ (input-only / output-intervention / strategy-structure). Distinct from the input set I.

$\mathrm{Adm}^{\mathfrak{F}}_M(\tau)$ — admissible counterfactual traces under the frame.

$\preceq_\tau$ — similarity order; canonically the count-first lexicographic *total* order (Section 4.2), unique minimum τ. $\uparrow_{\preceq_\tau}(X)$: traces *farther* from τ than some element of X.

$\mathrm{Close}^{\mathfrak{F}}_{\tau,M}(\neg C)$ — $\preceq_\tau$-minimal admissible $\neg C$-removals (over finite-change removals).

$\triangleleft$ — preference order on causes; canonically (size, depth, permissiveness), $C' \triangleleft C$ means C' preferred.

$\mathsf{Causes}_{M,\mathfrak{F},\tau}(E_{\mathfrak{F}})$ — valid causes (Section 4); $C^\star$ a $\triangleleft$-minimal one.

$C^\flat$ — the nearest-failure certificate (Section 4.7); determined only on $\mathrm{Adm}^{\mathfrak{F}}_M(\tau)$, generally not in $\mathcal{K}_{\mathfrak{F}}$.

N1 / N2 — trace-existential (one-player) / strategy-guarantee (two-player) necessity (Section 4.6).

$\Sigma_{\mathfrak{F}}, V_{\mathfrak{F}}$ — strategies agreeing with σ outside the editable vertex set $V_{\mathfrak{F}}$; D_C, T_C the causal dominion and its transitions.

$G^{\mathrm{rev}}_{M,\mathfrak{F},C,E}$, $W_\exists$, $\mathcal{A}[W_\exists]$ — the reverse parity game (Definition 3), its $\exists$-winning region, and the sub-arena restricted to it (Proposition 3).

A_{HOA}, Γ, C_Γ — TempoBench acceptor; an input-literal constraint set and its formula (Section 9).

at_s — atomic proposition exposing internal state s (prefix $at\_$ to avoid clash with input names).